

Multiplicative Langevin Process for Volatilities Produces Observed Q-Variance Regularities

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Abstract

Q-variance (so-called) posits a statistical relationship $\mathbf{E}(\sigma^2|z) = \sigma_0^2 + \frac{1}{2}z^2$ between an asset's volatility σ^2 , as observed in a time interval T , and its (suitably scaled) return z in the same interval. We here show that this relationship is *exactly equivalent* to positing an Inverse Gamma probability distribution for σ^2 itself. We then show that such a distribution is exactly generated by a multiplicative Langevin process with an arbitrary, settable coherence time τ_c , so that very nearly the same Q-variance relationship will hold for all $T \ll \tau_c$.

1 Introduction

Orrell [1, 2] has pointed out a surprisingly persistent regularity in asset price data, not explainable by a simple random walk model such as the now-long-standard Geometric Brownian motion (GBM) model [3]. If x (a random variable) denotes the logarithmic return over a time T , then, in a random walk model, the scaled $z = x/\sqrt{T}$ is a random variable that does not, under GBM, depend on T .

Now divide intervals of length T into n equal subintervals with respective scaled returns z_i , so that $\sum_i z_i = z$. Consider the subinterval variance $\sigma^2 \equiv \text{Var}(z_i)$. A property of the GBM model is that z and σ^2 are statistically independent (see Appendix A1).

Notwithstanding, Orrell finds, across a large corpus of asset price data, that σ^2 and z are not independent, and that, moreover, the conditional expectation $\mathbf{E}(\sigma^2|z)$, when measured by binning data by z values, empirically follows

$$\mathbf{E}(\sigma^2|z) = \sigma_0^2 + kz^2, \quad (1)$$

Here σ_0 is a parameter that depends on the specific asset, while k is observed to be close to the value $1/2$. Importantly, the parameters σ_0 and k for an asset are approximately independent of T . Orrell coins the term “Q-variance” for this phenomenon.

We see three distinct mysteries in the Q-variance regularities: (1) Why the observed universal functional form of equation (1)? (2) Why the observed independence of T . (3) Why the value $k = 1/2$?

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That observed price movement statistics are fat-tailed (that is, not Gaussian) has long been observed [4, 5, 6]. One large class of models explains this as the unfolding of a random walk not in clock time, but according to some latent variable, a “rate of information arrival”. These so-called subordinated time models [7, 8] make the probability distribution of price changes not Gaussian, but fat-tailed as a mixture of Gaussian distributions. Equivalently, one can regard an asset’s volatility σ^2 not as a constant, but as time varying $\sigma^2(t)$, the result of some stochastic process. Over long times, any such process, if stationary, implies some probability distribution $P(\sigma^2)$.

In the context of such models, this note looks at the implications of the three Q-variance mysteries. After defining some appropriate quantities in §2, we proceed in three steps: First (§3) we show that the Q-variance functional form of equation (1) uniquely implies that the marginal probability $P(z)$ must be a Student t-distribution. Second (§4) we show that this marginal $P(z)$ uniquely implies an Inverse Gamma distribution for the probability $P(\sigma^2)$. Third (§5) we show that this Inverse Gamma probability distribution is readily generated by a multiplicative Langevin process whose coherence time is adjustable to be longer than the largest T for which the Q-variance is seen to hold, thus making equation (1) approximately independent of T .

2 Conditional Probabilities and Expectations

We henceforth write $V \equiv \sigma^2$. The joint distribution $P(V, z)$ is given by

$$P(V, z) = P(z|V)P(V) \quad (2)$$

When $z \sim \mathcal{N}(0, V)$, we have

$$P(V, z) = \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{z^2}{2V}\right) P(V) \quad (3)$$

Bayes’ rule is

$$P(V|z) = \frac{P(z|V)P(V)}{P(z)} \quad (4)$$

where $P(z)$ is the marginal,

$$P(z) = \int_0^\infty \frac{P(V)}{\sqrt{2\pi V}} \exp\left(-\frac{z^2}{2V}\right) dV \quad (5)$$

so,

$$P(V|z) = \frac{V^{-1/2} P(V) \exp\left(-\frac{z^2}{2V}\right)}{\int_0^\infty V^{-1/2} P(V) \exp\left(-\frac{z^2}{2V}\right) dV} \quad (6)$$

The expectation of the variance V given a constant z is the first moment of the distribution, a ratio of integrals,

$$E[V|z] = \frac{\int_0^\infty V^{1/2} P(V) \exp\left(-\frac{z^2}{2V}\right) dV}{\int_0^\infty V^{-1/2} P(V) \exp\left(-\frac{z^2}{2V}\right) dV} \quad (7)$$

3 Q-variance Regularities Imply Student’s t-Distribution for the Marginal $P(z)$

Suppose that $E[V|z]$ is exactly $V_0 + kz^2$ (equation (1)). Then equations (5) and (7) imply

$$\int V \frac{1}{\sqrt{2\pi V}} e^{-z^2/2V} P(V) dV = (V_0 + kz^2) P(z) \quad (8)$$

Differentiate both sides with respect to z :

$$\int V \left(-\frac{z}{V} \right) P(z|V) P(V) dV = \frac{d}{dz} [(V_0 + kz^2) P(z)] \quad (9)$$

The left side is just $-zP(z)$, so,

$$-zP(z) = 2kzP(z) + (V_0 + kz^2) \frac{dP(z)}{dz} \quad (10)$$

Rearranging, we have

$$\frac{1}{P(z)} \frac{dP(z)}{dz} = -\frac{(1+2k)z}{V_0 + kz^2} \quad (11)$$

which can be integrated to give

$$P(z) \propto \left(1 + \frac{k}{V_0} z^2 \right)^{-\left(\frac{1}{2k}+1\right)} \quad (12)$$

which is Student's t-distribution with parameter $\nu = 1/k + 1$.

Equations (8)–(12) can be read in reverse order to prove the converse, that a Student t distribution for $P(z)$ implies the Q-variance regularity for $E[V|z]$.

4 Student's t Marginal Uniquely Implies Inverse Gamma Conditional

Here, we show that the Inverse Gamma distribution is the the unique mixing distribution that produces Student's t-distribution from a Gaussian scale mixture, hence is the unique distribution implied by the observed Q-variance functional form.

We seek to solve for $P(V)$ using the integral equation defined by the Gaussian scale mixture and the empirical marginal distribution (equations (5) and (12)),

$$\int_0^\infty \frac{1}{\sqrt{2\pi V}} \exp\left(-\frac{z^2}{2V}\right) P(V) dV \propto \left(1 + \frac{k}{V_0} z^2\right)^{-\left(\frac{1}{2k}+1\right)} \quad (13)$$

First, we change variables to turn the Gaussian kernel into a standard Laplace Transform. Let $s = z^2/2$ and $w = 1/V$. The differential becomes $dV = -(1/w^2)dw$. The limits $V \in (0, \infty)$ map to $w \in (\infty, 0)$. Substituting these into the integral, we obtain:

$$\int_0^\infty e^{-sw} \left[w^{-3/2} P(1/w) \right] dw \propto \left(1 + \frac{2k}{V_0} s\right)^{-\left(\frac{1}{2k}+1\right)} \quad (14)$$

The left side is exactly the Laplace Transform of the function $f(w) = w^{-3/2} P(1/w)$.

The inverse Laplace Transform of the right-hand side can be found using the standard table result,

$$\mathcal{L} \left\{ \frac{\beta^\alpha}{\Gamma(\alpha)} w^{\alpha-1} e^{-\beta w} \right\} = \left(1 + \frac{s}{\beta}\right)^{-\alpha} \quad (15)$$

yielding

$$\mathcal{L}^{-1} \left[\left(1 + \frac{2k}{V_0} s\right)^{-\left(\frac{1}{2k}+1\right)} \right] \propto w^{\frac{1}{2k}} \exp\left(-\frac{V_0}{2k} w\right) \quad (16)$$

Equating the Laplace transforms of the two sides give

$$w^{-3/2}P(1/w) \propto w^{\frac{1}{2k}} \exp\left(-\frac{V_0}{2k}w\right) \quad (17)$$

and replacing w with $1/V$,

$$P(V) \propto V^{-(\frac{1}{2k} + \frac{3}{2})} \exp\left(-\frac{V_0}{2kV}\right) \quad (18)$$

This shows that $P(V)$ is uniquely an Inverse Gamma distribution with parameters

$$\alpha = \frac{1}{2k} + \frac{1}{2} = \frac{1}{2}\nu, \quad \beta = \frac{V_0}{2k} \quad (19)$$

While this calculation does not indicate that the claimed value $k \approx 1/2$ should be special, it allows that value to be obtained by choosing the Inverse Gamma parameter $\alpha = 3/2$.

5 Stochastic Process Model for $\sigma^2(t)$

Thus far, we have “explained” the Q-variance functional form as deriving uniquely from an Inverse Gamma distribution for the probability distribution $P(\sigma^2)$. The value of the coefficient $k \approx 1/2$ is left as parameter deriving somehow from the psychology of markets, to be fitted from the data.

The remaining mystery, the invariance of Q-variance over a wide range of segment sizes T can be simply viewed as a statement that the stochastic process $\sigma^2(t)$ should have a coherency time $\gg T_{\max}$, the maximum T studied.

The challenge, then, is to exhibit a stochastic process whose stationary solution is Inverse Gamma distributed, but that also has an adjustable coherency time. That is straightforward. The Langevin process with multiplicative noise, [9, 10, 11]

$$dV_t = \gamma(\bar{V} - V_t)dt + sV_t dW_t \quad (20)$$

where γ is the mean-reversion rate, $\bar{V}$ is the target level, and s is the noise strength, is easily shown (see Appendix A2) to produce the stationary solution to the Fokker-Planck equation,

$$P(V) \propto V^{-(2+2\gamma/s^2)} \exp\left(-\frac{2\gamma\bar{V}}{s^2V}\right) \quad (21)$$

with a coherency time (see Appendix A3)

$$\tau_c \approx \frac{1}{\gamma + \frac{1}{2}s^2} \quad (22)$$

Mapping these to the Inverse Gamma parameters α and β gives

$$\alpha = 1 + \frac{2\gamma}{s^2}, \quad \beta = \frac{2\gamma\bar{V}}{s^2} \quad (23)$$

Alternatively, one can specify any desired values, for example, $k = \frac{1}{2}$ and $V_0 \equiv \sigma_0^2$, and any sufficiently large value τ_c , and then compute

$$\gamma = \frac{1-k}{(1+k)\tau_c}, \quad s^2 = \frac{4k}{(1+k)\tau_c}, \quad \bar{V} = \frac{V_0}{1-k} \quad (24)$$

The resulting multiplicative Langevin process equation (20) will then “explain” all three Q-variance mysteries: (1) $\mathbf{E}(\sigma^2|z) = \sigma_0^2 + kz^2$, (2) with σ_0^2, k as specified, and (3) invariant with T for $T \ll \tau_c$.

Positing a multiplicative Langevin process introduces a new parameter that can be taken as either γ or else s^2 separately, or else as the coherence time τ_c (equation (22)), and cannot be estimated from returns as an unordered set. As T approaches or exceeds τ_c , mean reversion will begin to pull the variance toward $\bar{V}$, causing $\mathbf{E}(\sigma^2|z)$ to plateau and driving k in equation (1) to zero. Any Gaussian mixture model thus implies that the Q-variance regularity cannot persist for all T , but must disappear as T gets large.

6 Relation to Previous Work

Stochastic models for time-varying variances, in both continuous and discrete time, have long been established in the finance literature [12, 13, 14]. Some widely used models, however, do not naturally yield the fat tails required to explain observed price statistics. For instance, the Heston model [6] employs an affine square-root process resulting in a Gamma stationary distribution for the variance, which lacks the heavy tails seen in empirical data. Other early models, such as Hull and White [12], did not incorporate mean reversion.

Nelson [15] notably demonstrated that the discrete-time GARCH(1,1) model converges to the multiplicative Langevin process (equation (20)) in the continuous limit. This process, often called Nelson’s diffusion, possesses both desirable properties: it is mean-reverting and yields a stationary Inverse Gamma distribution with power-law tails. Barone-Adesi et al. [16] have explored the implications of this framework for option pricing.

The contribution of this paper is to situate the “Q-variance” regularity observed by Orrell [1] within this established framework, providing a unified derivation that explains its functional form, its invariance over time scales T , and its characteristic coefficient.

7 Summary

The assertion (“Q-variance”) that $\mathbf{E}(\sigma^2|z) = \sigma_0^2 + kz^2$ holds with $k = 1/2$ is exactly equivalent to the statement that $P(\sigma^2)$ is Inverse Gamma distributed with $\alpha = 3/2$. Any other distribution $P(\sigma^2)$ will produce a different $\mathbf{E}(\sigma^2|z)$. Whether this value of k or α is universal, or is supported by any particular data set, is an issue to be discussed elsewhere.

The required $P(\sigma^2)$ is readily produced by a multiplicative Langevin process (also known as Nelson’s diffusion process) whose time coherence τ_c can be chosen to be much longer than any measured subinterval size T , producing arbitrarily good T independence for that and any smaller value of T .

Apart from σ_0^2 , which is observed to be different for different assets, the model has only one adjustable parameter, α or k , and arguably zero arbitrary choices of functions (unless one counts Langevin, equation (20), as arbitrary).

Disclosure of Use of AI

Gemini (“3 Fast”) and ChatGPT (GPT-5.3) assisted in developing and checking the equations in this paper. Each LLM checked the other’s version of the math, finding no significant discrepancies. We provided overall guidance and a final human check. The LLMs were not used in writing the manuscript. We are responsible for all errors in both the math and the words.

Appendices

A1. Random Walk Implies That z and σ^2 Are Statistically Independent

Let $\mathbf{1}$ be vector $(1, 1, 1, \dots, 1)$ of length n . Let $\mathbf{Z} = (z_1, z_2, \dots, z_n) \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ be the vector of scaled returns in the subintervals. Now decompose Z into vectors along and perpendicular to $\mathbf{1}$:

$$\begin{aligned} Z_{\parallel} &= (\bar{z}, \bar{z}, \dots, \bar{z}) \\ Z_{\perp} &= (z_1 - \bar{z}, z_2 - \bar{z}, \dots, z_n - \bar{z}) \end{aligned}$$

where $\bar{z} \equiv (1/n) \sum z_i$. Since these are orthogonal directions in the spherically symmetric normal distribution $\mathcal{N}(0, \sigma^2 \mathbf{1})$, they are statistically independent. Since the total return $z = Z_{\parallel} \cdot \mathbf{1}$ is a function only of $Z_{\parallel}$, while $\sigma^2 = \frac{1}{n-1} Z_{\perp} \cdot Z_{\perp}$ is a function only of the second, z and σ^2 must also be statistically independent.

A2. Multiplicative Langevin Process Yields Inverse Gamma

To get the Fokker-Planck stationary solution for the process

$$dV_t = \gamma(\bar{V} - V_t)dt + sV_t dW_t \quad (25)$$

we equate the drift and diffusion terms, so that there is no net flux of probability:

$$\gamma(\bar{V} - V)P(V) = \frac{1}{2} \frac{d}{dV} [s^2 V^2 P(V)] \quad (26)$$

Defining $G(V) = s^2 V^2 P(V)$, this becomes

$$\frac{\gamma(\bar{V} - V)}{s^2 V^2} G(V) = \frac{1}{2} \frac{dG}{dV} \quad (27)$$

whose solution is

$$G(V) \propto V^{-2\gamma/s^2} \exp\left(-\frac{2\gamma\bar{V}}{s^2 V}\right) \quad (28)$$

or

$$P(V) = \frac{G(V)}{s^2 V^2} \propto V^{-\left(\frac{2\gamma}{s^2} + 2\right)} \exp\left(-\frac{2\gamma\bar{V}}{s^2 V}\right) \quad (29)$$

which is an Inverse Gamma distribution with

$$\alpha = \frac{2\gamma}{s^2} + 1, \quad \beta = \frac{2\gamma\bar{V}}{s^2} \quad (30)$$

A3. Calculation of Coherency Time

We transform equation (25) into an Ornstein-Uhlenbeck (OU) process by changing variables to $x = \log V$ via Itô's Lemma. This yields

$$dx = \left[\gamma\bar{V}e^{-x} - \left(\gamma + \frac{1}{2}s^2 \right) \right] dt + s dW_t \equiv A(x)dt + s dW_t \quad (31)$$

The equilibrium is found by setting to zero the drift term, $A(x) = 0$, implying immediately

$$\gamma\bar{V}e^{-x} = \gamma + \frac{1}{2}s^2 \quad (32)$$

Near equilibrium, we can linearize the equations, in which case the relaxation rate λ (reciprocal of relaxation time τ_c) becomes the negative derivative of $A(x)$ at equilibrium,

$$\lambda = -(-\gamma\bar{V}e^{-x}) = \gamma + \frac{1}{2}s^2 \quad (33)$$

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